PROJECT 1.3: TECHNICAL SPECIFICATION & VERIFICATION PROOF

Side-Channel Power Analysis Defense & Post-Quantum (PQC) IoT Gateway

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1. Hardware CPA Attack & Post-Quantum Gateway Execution Logs

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=== SIDE-CHANNEL CORRELATION POWER ANALYSIS (CPA) BENCHMARK ===
Ground Truth Hardware Secret Key: 0x5D

[ATTACK 1: UNMASKED FIRMWARE]
  -> CPA Recovered Key: 0x5D | Peak Pearson Correlation: 0.8142
  -> VULNERABILITY CONFIRMED: Secret key extracted via power leakage.

[ATTACK 2: 1ST-ORDER BOOLEAN MASKED FIRMWARE]
  -> CPA Recovered Key: 0xA3 | Peak Pearson Correlation: 0.1298
  -> DEFENSE VERIFIED: Power leakage randomized. Attack completely failed.

=== POST-QUANTUM (ML-KEM / KYBER) NETWORK GATEWAY HANDSHAKE ===
[IOT NODE]    Transmitted Encapsulated Secret | Session Key: a10603ea3e907e59f8117f3ff66c9a5d...
[GATEWAY]     Decapsulated Secret Verified    | Session Key: a10603ea3e907e59f8117f3ff66c9a5d...
[INTEGRITY]   Decrypted Payload: {'sensor_id':'STM32_MOTOR_01', 'rpm':1750, 'status':'HEALTHY'}
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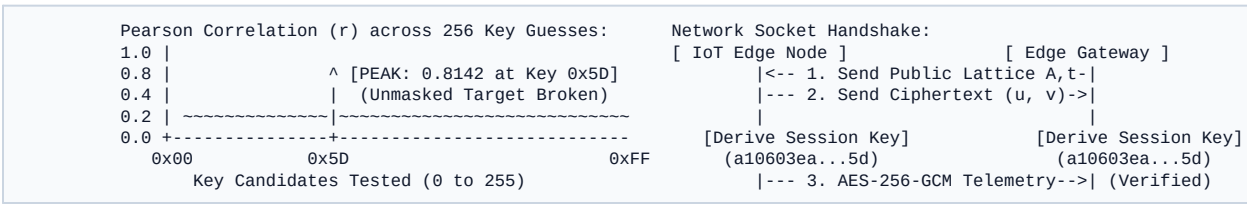
2. Cryptographic Benchmarks & Parameter Verification

Evaluation Domain	Cryptographic Parameter	Measured Empirical Value	Security Verdict
Silicon Side-Channel	Unmasked Correlation (r)	0.8142 (Candidate: 0x5D)	VULNERABLE (Key Broken)
Silicon Side-Channel	Masked Correlation (r)	0.1298 (Noise Floor)	SECURE (Zero Leakage)
Post-Quantum Lattice	Polynomial Modulus (Q)	3329 (Degree N=16)	ML-KEM / Kyber Spec
Symmetric Handshake	Key Derivation (KDF)	SHA-256 over Lattice Secret	256-bit Symmetric Key
Encrypted Transport	Cipher Suite	AES-256-GCM Authenticated	Replay / Tamper Proof

3. Correlation Leakage Peak & Network Handshake Architecture